

A150 – Quarks and Neutrinos: Status and Speculative Fraction

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Contents

.1	Purpose: Marking the Boundary	1
.2	Quarks	1
	Up-Type Quarks: Geometric Ladder at [K] Level	1
	Down-Type Quarks: Isospin Step Size	1
.3	Neutrinos	3
.4	What the Sector Still Achieves	3
.5	Verification Script	4
.6	Open Questions	4
.7	Supersedes	4
.8	Use of AI Tools	5

.1 Purpose: Marking the Boundary

The leptons are treated in A100 to A120, with clear layers and clear tolerances. For quarks and neutrinos the situation is different, and this document says above all *how* different.

[STIPULATION] This sector has two layers: the up-type quark ladder is at [K] level (geometric exponent sequence, numerically verified). Down-type quarks and neutrinos are [S]: sketch, candidate, plausibility. There is no statement there of the rank of the Koide relation or the up-type ladder.

.2 Quarks

[STIPULATION] The quark sector has two different layers that must be kept separate: the up-type ladder (u, c, t) is verified at [K] level; down-type (d, s, b) and neutrinos remain [S].

Up-Type Quarks: Geometric Ladder at [K] Level

[K] The up-type quarks follow the same geometric sequence as the leptons – exponent $p_{n+1} = (2/3) \cdot p_n$, starting value $p_u = 3/2$:

$$p_u = \frac{3}{2}, \quad p_c = \frac{2}{3}, \quad p_t = -\frac{1}{3}.$$

The prefactors r_i follow from the experimental masses; the exponents are geometrically determined. Numerical verification:

Quark	p_i	r_i	m_{FFGFT}	Exp. (PDG)
Up	+3/2	5.703	2.16 MeV	2.16 MeV
Charm	+2/3	1.978	1270 MeV	1270 MeV
Top	-1/3	1/28	172.0 GeV	172.76 GeV

Top quark deviation: 0.45 %. The top is the only fermion with a negative exponent – inverse mode rather than suppression, $\xi^{-1/3} \approx 19.6$ instead of ξ^{+p} .

[K] Where the ladder has [K] level and where not. The p_i values follow geometrically; that is shown. The r_i prefactors are back-calculated from the masses – they do not have the same [K] level as the r_i of the leptons, for which the circulant and the Koide relation give an independent structural justification. Open: why precisely these rungs are occupied (u, c, t) and not others.

Down-Type Quarks: Isospin Step Size

[K] The down-type quarks follow an arithmetic sequence:

$$p(n) = \frac{3}{2} - n \cdot \frac{1}{2}, \quad n = 0, 1, 2 \Rightarrow (p_d, p_s, p_b) = \left(\frac{3}{2}, 1, \frac{1}{2}\right).$$

Numerically (all deviations < 1 %):

Quark	p_i	r_i	m_{FFGFT}	Exp. (PDG)
Down	3/2	12.33	4.67 MeV	4.67 MeV
Strange	1	2.85	93.4 MeV	93.4 MeV
Bottom	1/2	1.47	4.18 GeV	4.18 GeV

[B] Derivation of the step size from torus geometry and isospin. The down quarks are the only fermion group with a uniform d_{akt} step per generation:

$$\Delta d_{\text{akt}}(\text{Down}) = -\frac{3}{2} \quad \left(-\frac{3}{2}, -1 \text{ leptons}; -\frac{5}{2}, -3 \text{ up-quarks}\right).$$

The uniformity explains the arithmetic p -sequence. The step $-3/2 = -D_{\text{space}} \cdot |T_3|$ follows from two components:

1. $D_{\text{space}} = 3$: spatial dimensions after decompactification $T^4 \rightarrow \mathbb{R}^3 \times \mathbb{R}_t$ (A090).
2. $T_3(\text{Down}) = -1/2$: weak isospin in the $SU(2)_L$ doublet (A192).

Thus:

$$\Delta p = \frac{\Delta d_{\text{akt}}}{D_{\text{space}}} = \frac{-D_{\text{space}} \cdot |T_3|}{D_{\text{space}}} = -|T_3| = -\frac{1}{2}.$$

Steps 1, 2 and the division are **[B]**. What remains **[S]**: the connection rule " $\Delta d_{\text{akt}} = -D_{\text{space}} \cdot |T_3|$ per generation" is geometrically plausible (each generation loses a T_3 -weighted spatial dimension of active mode), but has not yet been formally derived from the projection geometry.

Why this is worth less than the up-type ladder – three reasons:

First, quark masses are not directly measured quantities. They are scheme-dependent and run with the energy scale. A percent-level comparison presupposes fixing scheme and scale – and this choice is itself a degree of freedom.

Second, there are six quarks instead of three leptons, but no ordering corresponding to the Z_3 structure that would lead them to three eigenvalues of an operator. The circulant layer from A110 has no counterpart here.

Third, the number of structural numbers to be adjusted is correspondingly larger. By A020, a ξ -power is a statement only when its exponent is independently fixed. For the quarks this is not shown.

[STIPULATION] The A-series therefore treats the quark sector as a descriptive classification, not as a result, and does not count it among the prediction quantities.

[S] The sharpest individual case: the sign of the top quark. A concrete example shows where the boundary lies exactly. The top quark carries in the ladder representation a *negative* exponent $p_t = -\frac{1}{3}$, which gives an amplification $\xi^{-1/3} \approx 19.6$ instead of a suppression – the top is the only fermion that sits not below but at the electroweak scale (its Yukawa coupling is ≈ 1 , all others are $\ll 1$). So far the reliable observation.

[S] The sign is structured, not arbitrary. The exponents are rungs of a logarithmic resonance ladder with fixed step size $\Delta p = 1/3$; each step corresponds to the geometric factor $\xi^{1/3} \approx 1/19.6$. This ladder follows from $p = d_{akt}/3$, the number of active torus dimensions – a connection that is *linear* in d_{akt} and therefore admits negative rungs without difficulty. In this sense $-\frac{1}{3}$ is not an invented number but the rung one unit *below* the Higgs scale ($p = 0$), the opposite direction of the $+\frac{1}{3}$ step. A common short-circuit interprets the same instead as a “counter-winding” via the Laplace operator ($\lambda = n_1^2 + n_2^2 + n_3^2$); this does not hold because this operator is *even* in each winding number ($n = -1$ and $n = +1$ give the same eigenvalue) and thus cannot generate any sign. The supportable justification is the linear dimension ladder, not the counter-winding.

[STIPULATION] If $\frac{1}{3}$ is only the *step size* of the ladder, then $-\frac{1}{3}$ is simply its continuation one step *below* the Higgs reference ($p = 0$) – not a mirror that would need a partner at $+\frac{1}{3}$, but the next regular position downward. The occupation of individual rungs is a different question from their existence. Noteworthy is that even the bottom at $p = \frac{1}{2}$, i.e. $d_{akt} = 1.5$, occupies a *non-integer* rung: the interpretation of d_{akt} as a number of active dimensions therefore does not even hold throughout the positive range. Primary is the resonance axis with step $\frac{1}{3}$; the “dimensions” are only an illustration for the integer rungs. On this axis $-\frac{1}{3}$ is a regular position. What remains open is therefore not the *sign* but why the top is on precisely this rung and which rungs are occupied at all – the assignment, not the scale. The value therefore remains on the [S] layer: a regular position on a justified axis, whose occupation has not yet been derived.

.3 Neutrinos

[S] The neutrino sector is the weakest of the entire corpus, and that should be stated clearly here.

The absolute masses are experimentally unknown, only differences of mass squares. The mass ordering is open. Whether these are Dirac or Majorana particles is open. A theory that delivers numbers into this situation can hardly be refuted by the data – and what cannot be refuted confirms nothing either.

[S] The view: neutrinos as nearly massless photons. FPGFT treats the neutrino not like the charged leptons but by analogy with the photon. The photon is exactly massless,

$$\text{Photon: } E^2 = (pc)^2 + 0,$$

the neutrino carries a tiny residual mass from a *double* ξ suppression,

$$\text{Neutrino: } E^2 = (pc)^2 + \left(\sqrt{\frac{\xi^2}{2}} mc^2\right)^2.$$

The first ξ factor makes the particle “nearly massless” like a photon, the second leaves a small mass. As an estimate this gives

$$m_\nu = \frac{\xi^2}{2} m_e = \frac{(4/3 \cdot 10^{-4})^2}{2} \cdot 0.511 \text{ MeV} \approx 4.5 \text{ meV}.$$

[S] What the estimate achieves and what it does not. It captures the *qualitative* character: the neutrino is quasi-massless and lies many orders of magnitude below the charged

leptons – the photon analogy explains *why* this is so without a free mass parameter. But it does not hit the *scale*. The oscillation data require model-independently that the heaviest neutrino weighs at least $\sqrt{\Delta m_{\text{atm}}^2} \approx 50$ meV; the estimate gives only ~ 5 meV, roughly one order of magnitude too small (sum ~ 14 meV against the required ≥ 60 meV: factor ~ 4). The approach thus explains the *smallness*, not the *magnitude* of the neutrino mass – it is a picture with the right direction and the wrong scale, and remains explicitly on the [S] layer.

[STIPULATION] The sector is therefore tracked but not weighted. It belongs to the points where the theory has not yet reached its range.

.4 What the Sector Still Achieves

[B] One statement remains supportable and is independent of the numerical values: the generation number three follows from the Z_3 structure (A030, A110) and applies to all fermions, not only the leptons. That there are three quark families and three neutrino species need not be assumed additionally in this structure.

This is a structural statement without free parameters. It is weaker than a statement about numerical values – and it is one of the two reliable things this sector offers. It does not belong to the four prediction quantities of the series.

[B] The second reliable statement concerns not the masses but the *mixing*, and it is also parameter-free. FFGFT separates a magnitude from a phase sector (A120): the magnitude is the mass spectrum, the phase the mixing. In this reading the most striking difference between quarks and neutrinos is precisely a *magnitude-versus-phase* contrast:

- **Quarks:** strong mass hierarchy (large magnitude), small mixing – the CKM matrix is close to the identity.
- **Neutrinos:** flat, nearly degenerate magnitude, large mixing – the PMNS matrix is far from the identity.

This can be quantified. Measuring the off-diagonal weight of the $|U|^2$ matrix (per row) gives

$$\text{CKM: } 0.035 \quad \text{versus} \quad \text{PMNS: } 0.554,$$

a factor ~ 16 . The famous contrast “small versus large mixing” is thus structurally exactly the contrast “magnitude versus phase” – one family sits in the magnitude sector, the other in the phase sector. This statement uses no derived masses and no adjusted exponent; it only reads the measured mixing contrast into the magnitude/phase separation that FFGFT already carries.

[STIPULATION] This contrast is also not one of the four prediction quantities. It is a classification of the known, not a new numerical value – but one that makes do without free parameters and therefore stands above the [S] layer of the rest of the sector.

.5 Verification Script

- The CKM/PMNS contrast from measured mixing matrices: off-diagonal weight 0.035 versus 0.554, factor ~ 16 – without derived masses, without adjusted exponents: `python/A_Serie_Skripte/a150_quarks_neutrinos.py`

.6 Open Questions

First, the quark exponents are not independently justified.

Second, the neutrino sector is largely not testable – the absolute masses, the ordering, and the Dirac/Majorana character remain open. The one point that is testable goes against the theory: the ξ^2 estimate lies an order of magnitude below the scale required by oscillations. This does not disqualify the sector – an estimate is not a model – but it shows that no supportable mass scale yet stands here.

Third, a counterpart to the circulant layer is missing. Without one, the sector remains at the accuracy level of the ladder, i.e. at percent level, and nothing lies below that.

.7 Supersedes

This document subsumes: Doc. 005 (quarks), Doc. 007 (neutrinos, photon analogy), Doc. 047 (masses), Doc. 289 (magnitude/phase, CKM/PMNS contrast), Doc. 189 (torus windings, sign picture), Doc. 202 and Doc. 006 (top special case $p_t = -1/3$ as inverted hierarchy without derivation). Incorporated: P35, P40.

.8 Use of AI Tools

This document was created with the assistance of AI tools. The questions, stipulations, and all substantive decisions come from the author; the tools formulate, compute, and create verification scripts. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.